

1. Problem 1 (9-2)

- (a) The following equipment needs pressure relief: the 2 PD pumps, the 2 heat exchangers, the reboiler, the still, and the drum.
- (b) Relief scenarios:
 - PD pumps: outlet is blocked in, liquid service.
 - Heat exchanger: outlet is blocked in, liquid service.
 - Reboiler: high temperature leading to overpressure, vapor service.
 - Still: high temperature leading to overpressure, vapor service.
 - Drum: high level leading to overfill or overpressure, liquid service.
- (c) Type of relief:
 - PD pumps: spring valve.
 - Heat exchanger: spring valve.
 - Reboiler: rupture disk followed by spring valve.
 - Still: rupture disk followed by spring valve.
 - Drum: rupture disk followed by spring valve.

2. Problem 2 (10-2)

$$P_o = \frac{5}{50} = 10\%$$

$$K_p = 0.2533 + 0.03632P_o$$

$$K_p = 0.2533 + 0.03632 \cdot 10 = 0.6165$$

$$A = \frac{1}{38} \cdot \frac{Q_v}{C_0 K_v K_p K_b} \sqrt{\frac{\rho/\rho_{ref}}{1.25P_s - P_b}}$$

$$A = \frac{1}{38} \cdot \frac{100}{0.65 \cdot 1 \cdot 1 \cdot 0.6165} \sqrt{\frac{1}{1.25 \cdot 50 - 5}}$$

$$A = 0.866$$

$$D = \sqrt{\frac{4A}{\pi}}$$

$$D = \sqrt{\frac{4 \cdot 0.866}{\pi}}$$

$$\boxed{D = 1.05 \text{ in}}$$

3. Problem 3 (10-3)

$$\xi = 519.5 \sqrt{\gamma \left(\frac{2}{\gamma + 1} \right)^{(\gamma+1)/(\gamma-1)}}$$

$$\xi = 519.5 \sqrt{1.3 \left(\frac{2}{1.3 + 1} \right)^{(1.3+1)/(1.3-1)}}$$

$$\xi = 346.6$$

$$P_b = 17.7\%$$

$$K_b = 1$$

$$P = 100 \cdot 1.1 + 14.7 = 124.7$$

$$A = \frac{Q_m}{C_0 \xi K_b P} \sqrt{\frac{Tz}{M}}$$

$$A = \frac{50}{0.975 \cdot 346.6 \cdot 1 \cdot 124.7} \sqrt{\frac{559.7 \cdot 1}{28}}$$

$$A = 0.005305$$

$$D = \sqrt{\frac{4 \cdot 0.005305}{\pi}}$$

$$\boxed{D = 0.08218 \text{ in}}$$

4. Problem 4 (10-4)

$$\Delta P = 100 - 5 + 10 = 105$$

$$A = \frac{Q_v}{38C_0} \sqrt{\frac{\rho/\rho_{ref}}{\Delta P}}$$

$$A = \frac{500}{38 \cdot 0.61} \sqrt{\frac{1.2}{105}}$$

$$A = 2.31$$

$$D = \sqrt{\frac{4 \cdot 2.31}{\pi}}$$

$$\boxed{D = 1.71 \text{ in}}$$

5. Problem 5 (10-5)

$$\xi = 519.5 \sqrt{1.4 \left(\frac{2}{1.4 + 1} \right)^{(1.4+1)/(1.4-1)}}$$

$$\xi = 355.7$$

$$P = 100 \cdot 1.1 + 14.7 = 124.7$$

$$A = \frac{Q_m}{\xi P} \sqrt{\frac{Tz}{M}}$$

$$A = \frac{100}{355.7 \cdot 124.7} \sqrt{\frac{559.7 \cdot 1}{28}}$$

$$A = 0.01$$

$$D = \sqrt{\frac{4 \cdot 0.01}{\pi}}$$

$$D = 0.113 \text{ in}$$